Writing the SUPER Bootloader

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The bootloader of the factory mainboard is the version corresponding to JetPack 6.2, while the provided SSD uses the JetPack 7.2.1 system. Before first use, the mainboard bootloader needs to be updated to **L4T R39.2.1**, and then the provided SSD is installed for booting.

The writing process only updates the bootloader and UEFI in the mainboard QSPI, and does not write the SSD system. **No SSD needs to be installed during writing; it is recommended to remove the SSD with the power off first.**

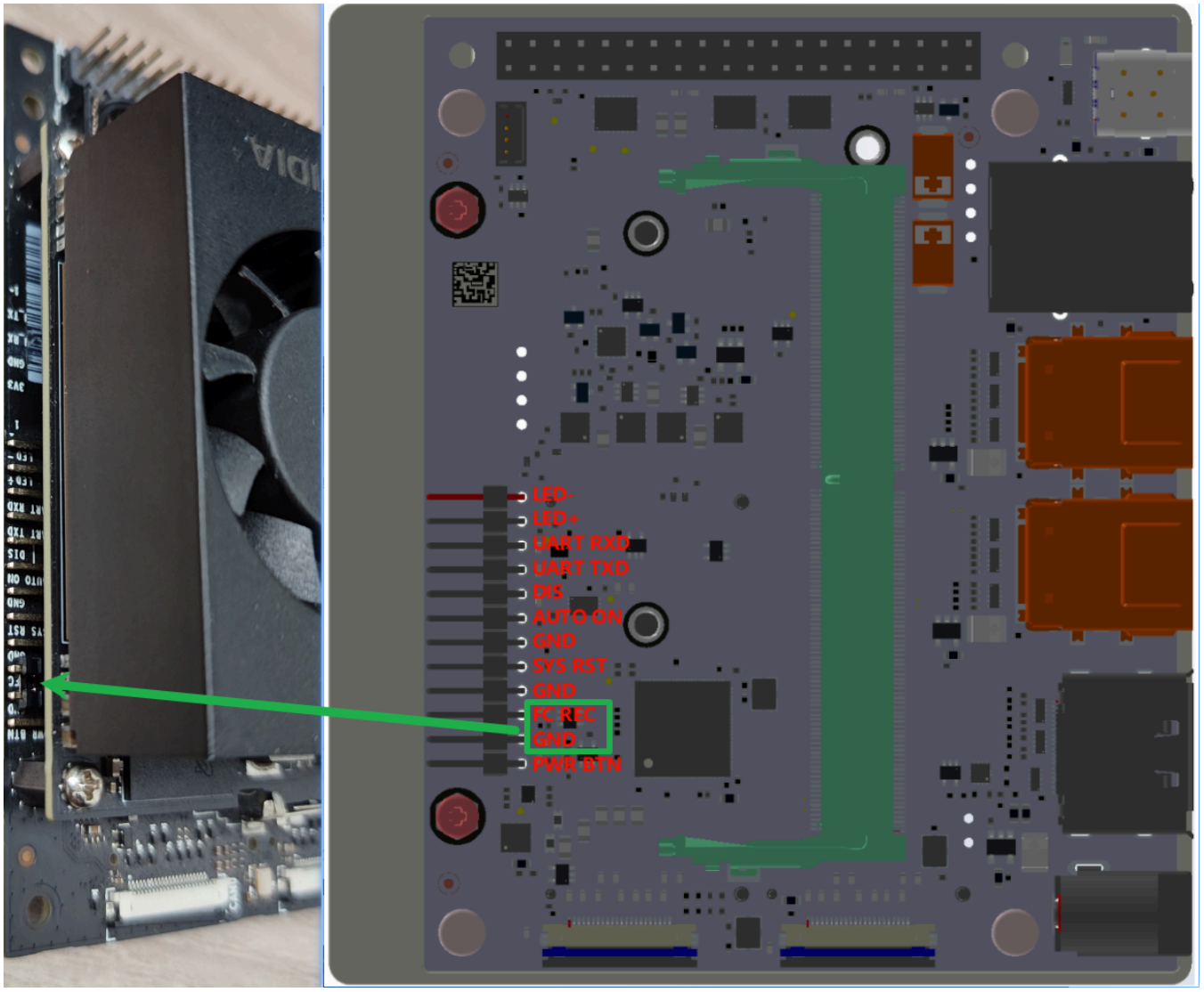
This tutorial follows the instructions for the NVIDIA official Orin Nano Developer Kit with the Super configuration. For writing, the virtual machine provided in the cloud drive or your own virtual machine system can be used; either Ubuntu 22.04 or 24.04 is acceptable.

> Official writing link: https://docs.nvidia.com/jetson/archives/r39.2.1/DeveloperGuide/SD/FlashingSupport.html#qspi-only-flashing

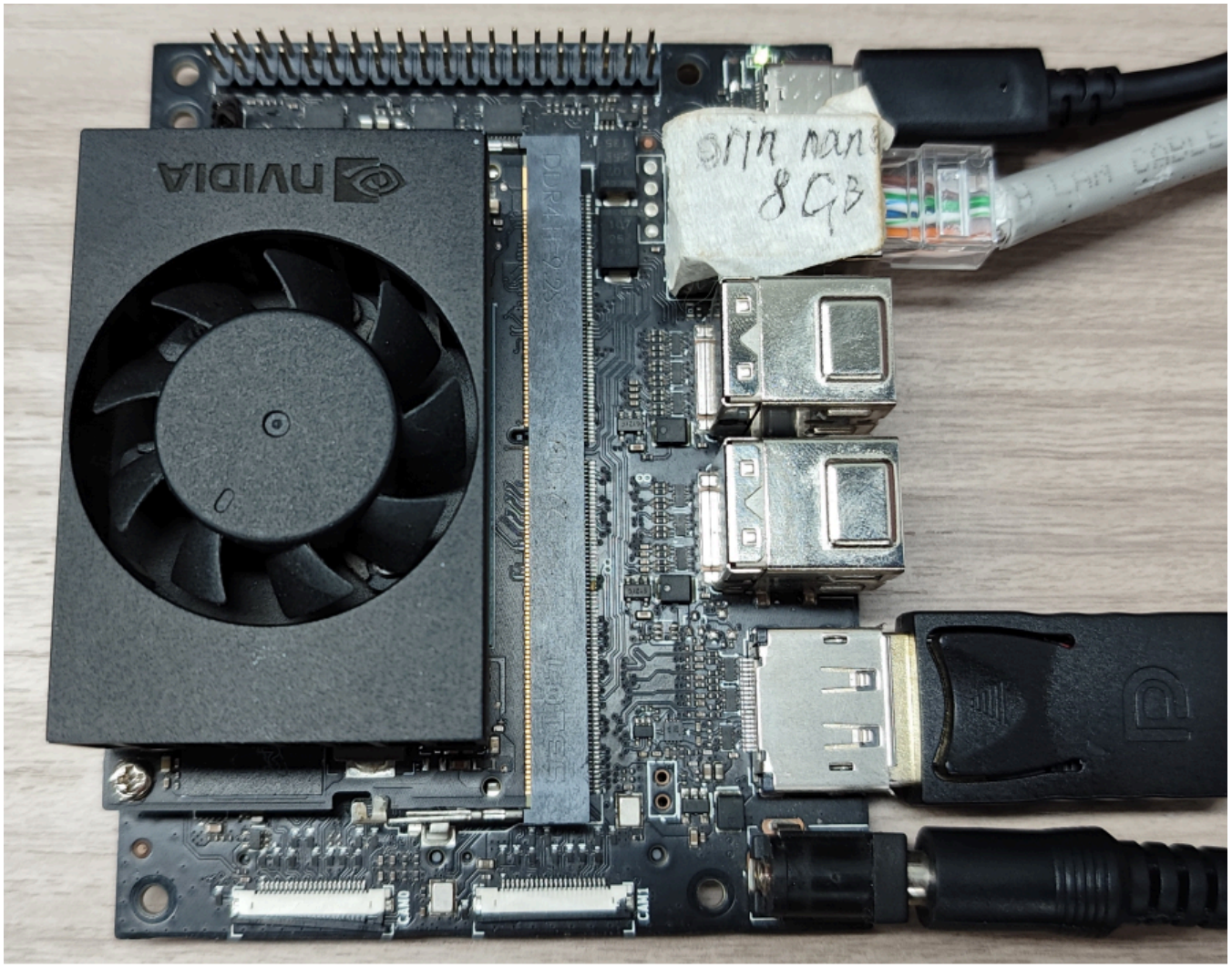
1. Recovery Mode

1.1. Hardware Connection

1. Disconnect the Jetson mainboard power and remove the SSD.
2. According to the pin markings on the carrier board, use a jumper cap to short `FC REC` and `GND`. The relevant pins of the official Orin Nano carrier board are located under the core board, and the operation can be performed without removing the core board.



3. Use a Type-C data cable that supports data transfer to connect the Jetson mainboard and the computer, then connect the DC power adapter to power the mainboard so that it enters Recovery mode.

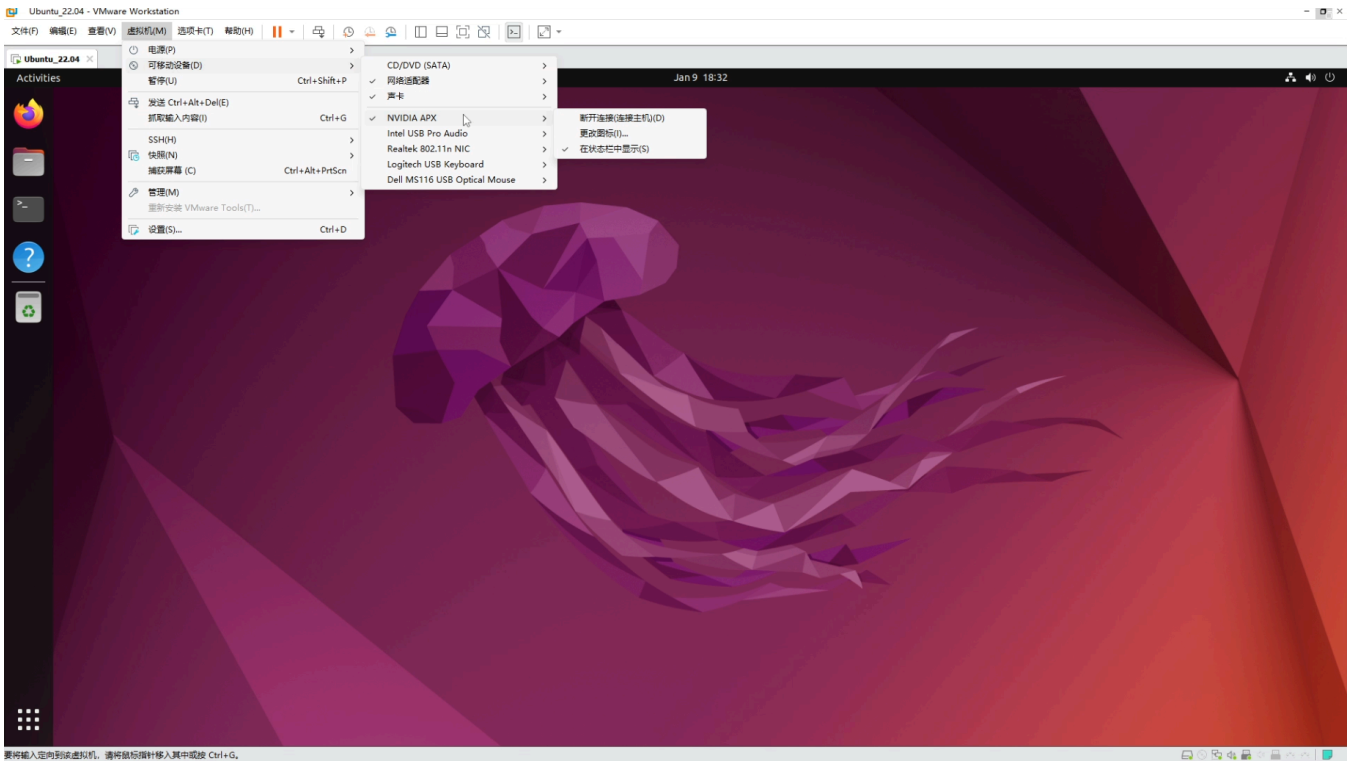


Note: Writing the bootloader does not require a DP data cable or a network cable; a monitor needs to be connected later when checking the boot screen. Keep the mainboard powered and the data cable connected during writing.

1.2. Software Connection

Start the Ubuntu virtual machine, find the NVIDIA USB device in "Virtual Machine → Removable Devices" of VMware, and connect it to the virtual machine.

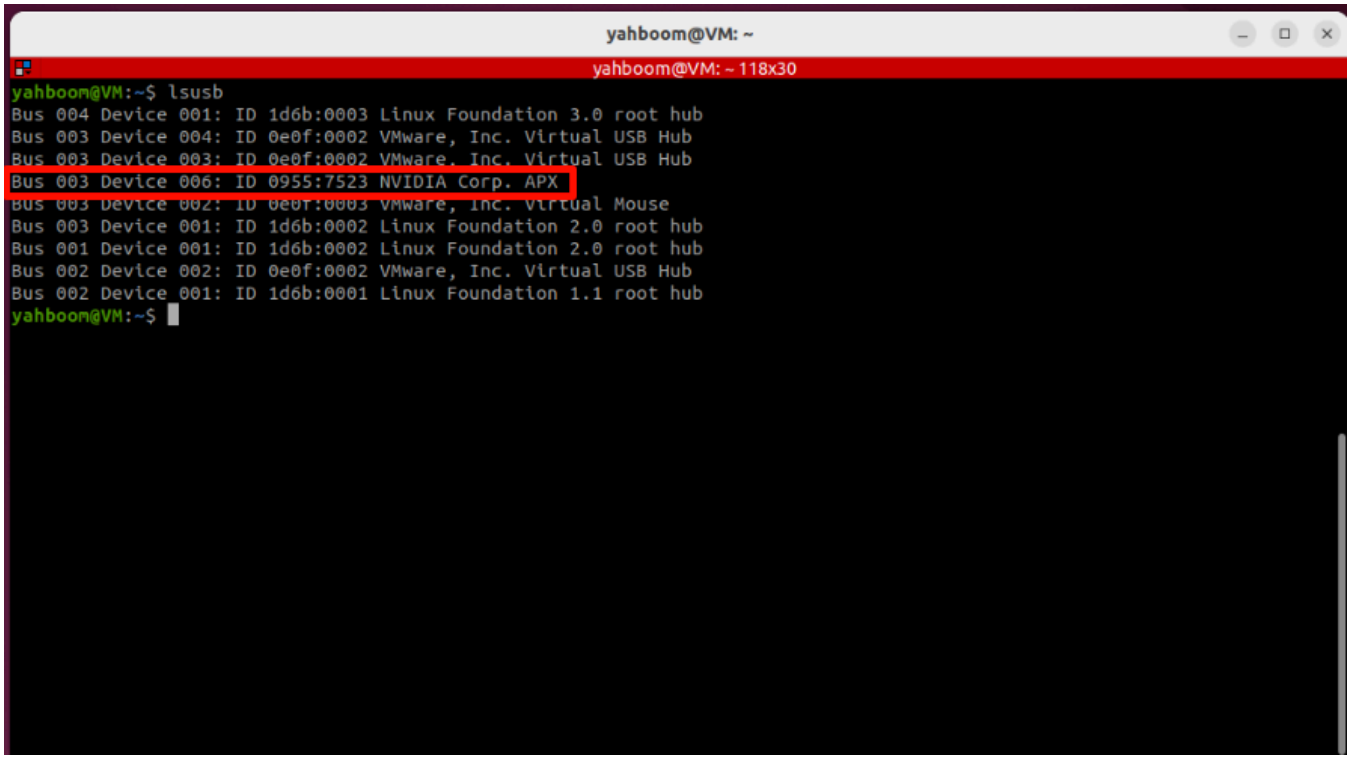
The device name may be displayed as `NVIDIA Corp. APX`. After selecting Connect, the device should be used by the Ubuntu virtual machine.



Open a terminal on the virtual machine desktop and run:

```
lsusb
```

Confirm that the NVIDIA device appears in the list with the vendor ID `0955`. If it is not recognized, first check the jumper cap, the power-on sequence, the Type-C data cable, and the VMware USB connection status, and continue only after confirming.



2. Writing the Bootloader

Open a terminal as a normal user in the virtual machine. The username and password of the provided virtual machine are both `yahboom`; if your own Ubuntu system is used, use your own account and confirm that the writing dependencies are installed.

Enter the folder and confirm that the writing script exists:

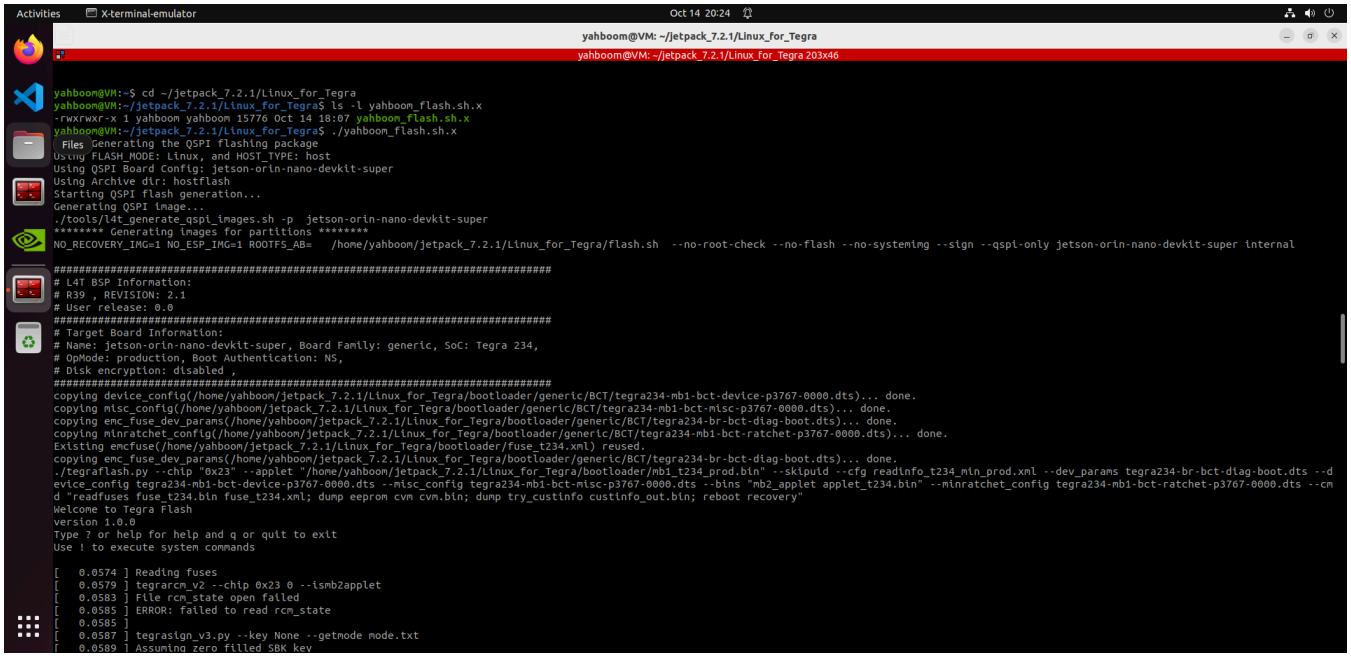
```
cd ~/jetpack_7.2.1/Linux_for_Tegra
ls -l yahboom_flash.sh.x
```

Then run:

```
./yahboom_flash.sh.x
```

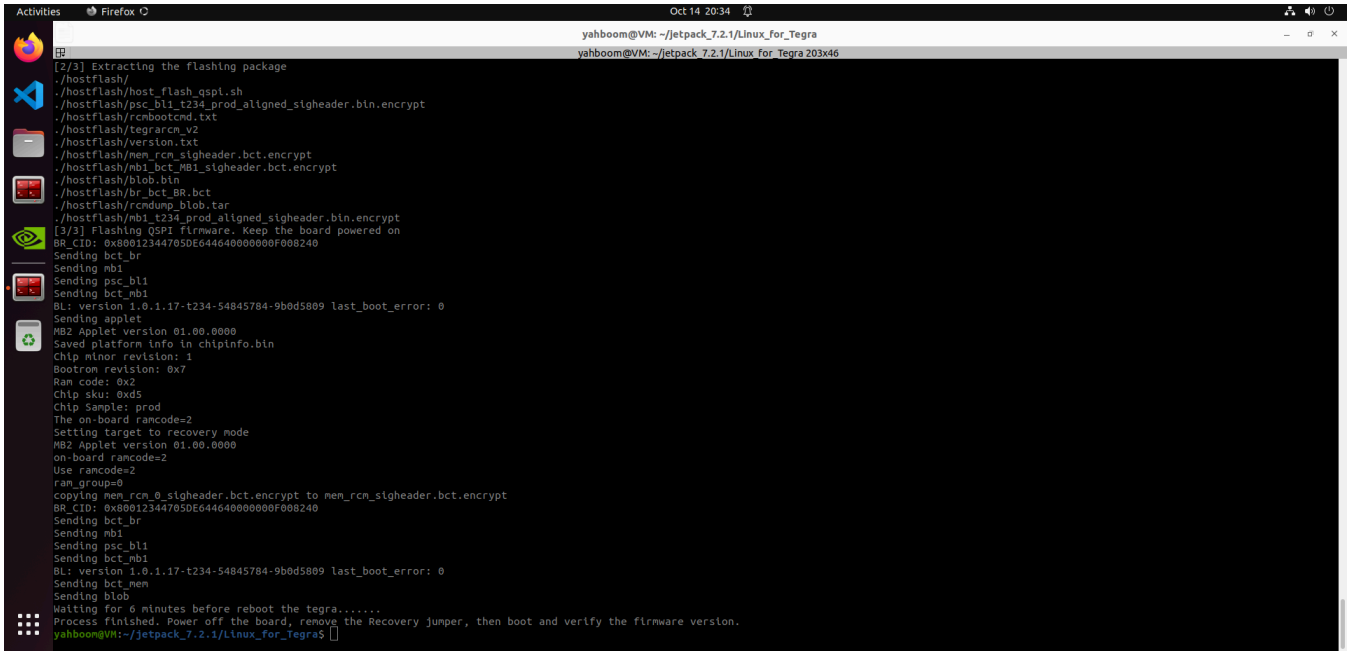
The script performs the following in sequence:

1. Generate the QSPI bootloader writing package for R39.2.1.
2. Decompress the generated writing package.
3. Send the temporary writing system to the mainboard, and the mainboard completes the bootloader writing.



After the terminal displays the following content, keep the mainboard powered and wait about **6 minutes**:

```
Sending bct_mem
Sending blob
Waiting for 6 minutes before reboot the tegra.......
```



Note: At this stage, the NVIDIA APX device will disconnect, and the motherboard fan may begin to spin. This is a normal mode transition that occurs when the temporary flashing system takes control; it is not required for the APX connection to remain active. Do not power off the device, force it back into Recovery mode, or repeatedly reconnect the APX in response to this.

The terminal will not display new logs while you wait, and the script used here is not configured to automatically reboot the motherboard. Once the process concludes and the script returns to the command prompt, proceed to the instructions in Section 3. If the script encounters an error, please review the error messages provided above. Ensure the process has fully stopped before powering off the device; then, follow the steps in Section 1 to re-enter Recovery mode and reconnect the virtual machine. Never power off the device while data is still being written to the board.

> If you are using your own virtual machine and encounter a flashing error after running the commands, please verify whether the QEMU user-mode emulation tools are installed. If they are missing, run the following command to install them:

```
sudo apt-get update
sudo apt-get install -y qemu-user-static
```

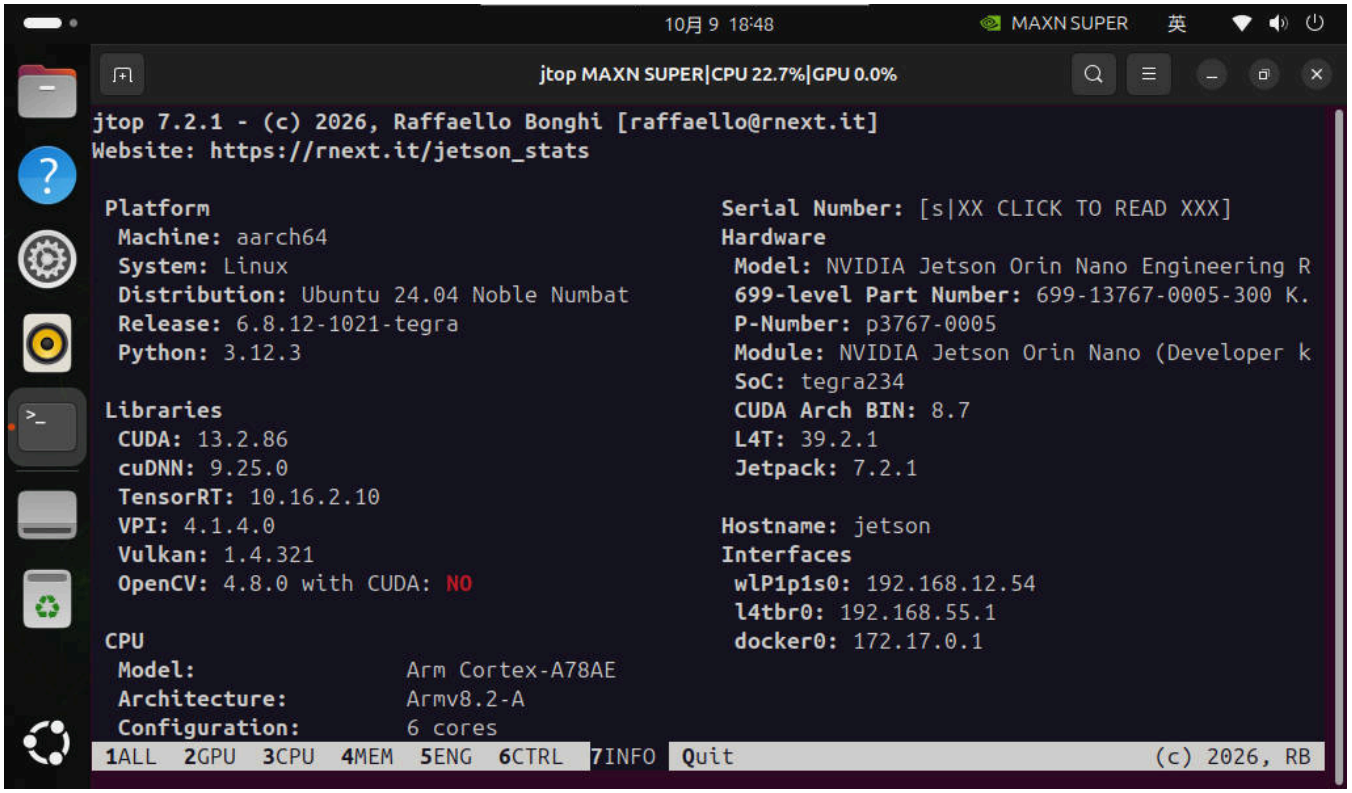
3. Starting the System

1. After the writing script ends, if a debug serial port is connected, first confirm that the writing on the board is complete. Then disconnect the Jetson power and the Type-C data cable.
2. Remove the Recovery jumper cap shorting `FC REC` and `GND`.
3. Install the provided JetPack 7.2.1 SSD and connect a monitor.
4. Reconnect the mainboard power and observe the UEFI firmware version on the boot screen; it should display `39.2.1`.



If `36.x` is still displayed, this bootloader update did not take effect. Check that you used the correct provided files and a freshly generated writing package, and troubleshoot with the board logs.

After entering the system, open a terminal and run jtop to view the information



After completion, reboot normally once and confirm that the mainboard can boot from the provided SSD; the subsequent tutorials can then be used.